

WELL COMPLETION REPORT (SYNTHETIC)

OIL-RDS-011 - Rudrasagar-011

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000126
Field / District	Rudrasagar / Sivasagar
Block	PEL-ASSAM-06
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2010-12-03 / 2011-04-16
Measured / TVD	2559 m / 2529 m
Trajectory	Vertical; inc 2.2 deg; az 157.0 deg
Formation Interval	1505-2099 m
Lithology	Fluvial sandstone with clay streaks
Pore Pressure Reference	2773 psi
Fracture Gradient	14.86 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (48.5/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	639	Surface
9 5/8 in casing	1484	Intermediate
7 in liner/casing	2251	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2010-12-22 495 m	Stuck Pipe	Critical	Hookload became erratic during pipe movement. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2011-04-08 2417 m	Cementing Issue	Critical	Displacement pressure became unstable. Mitigation: Review loss history before cementing equivalent interval. Lesson: Plan conservative displacement rate in weak formations.
2010-12-17 391 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Use offset torque trend to set adaptive operating envelope.
2010-12-27 598 m	Fishing Operation	High	Mechanical obstruction prevented normal drilling progression. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2010-12-29 638 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2011-01-21 1069 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2011-02-15 1511 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.

Completion Summary

The synthetic well reached 2559 m MD in the Tipam interval. The NWIS knowledge record links 18 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.